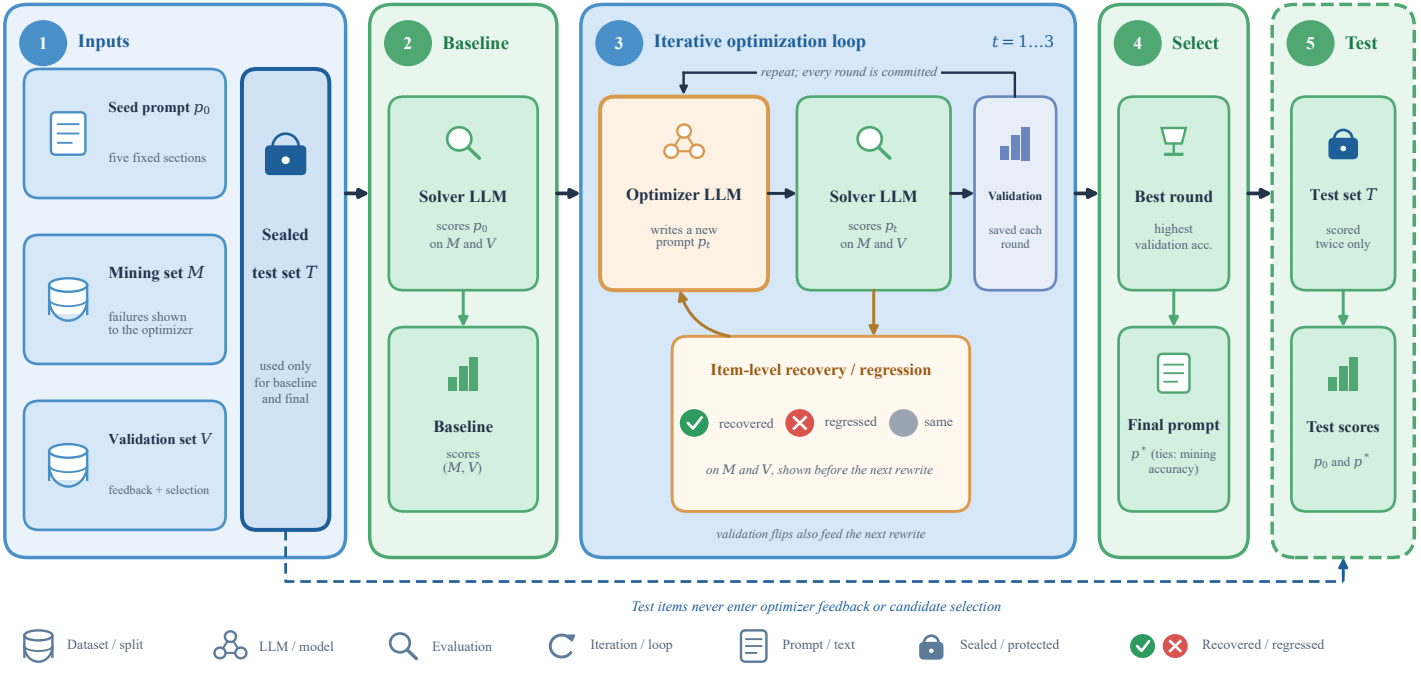

1
Inputs
Seed prompt p_0
five fixed sections
Mining set M
failures shown to the optimizer
Validation set V
feedback + selection
Sealed test set T
used only for baseline and final
2
Baseline
Solver LLM
scores p_0 on M and V
Baseline
scores (M, V)
3
Iterative optimization loop
$t = 1...3$
repeat; every round is committed
Optimizer LLM
writes a new prompt p_t
Solver LLM
scores p_t on M and V
Validation
saved each round
Item-level recovery / regression
recovered
regressed
same
on M and V, shown before the next rewrite
validation flips also feed the next rewrite
4
Select
Best round
highest validation acc.
Final prompt
p^* (ties: mining accuracy)
5
Test
Test set T
scored twice only
Test scores
p_0 and p^*
Test items never enter optimizer feedback or candidate selection
Dataset / split
LLM / model
Evaluation
Iteration / loop
Prompt / text
Sealed / protected
Recovered / regressed